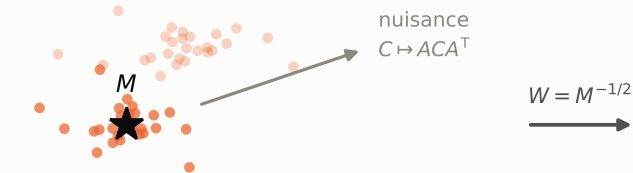


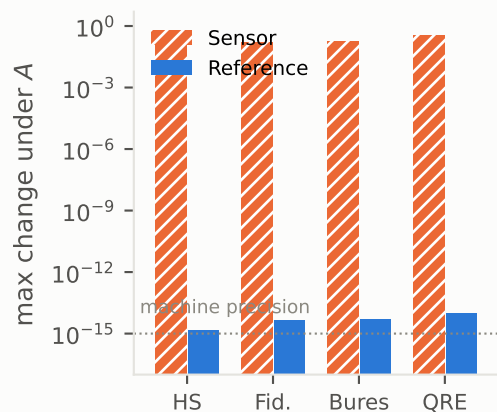
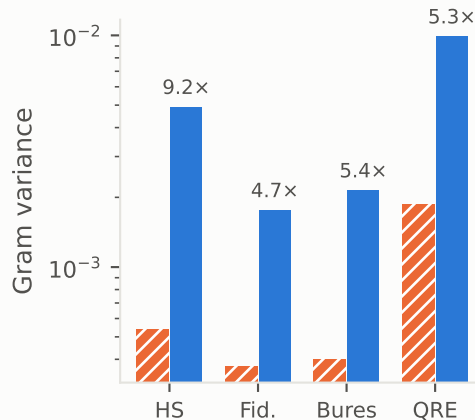
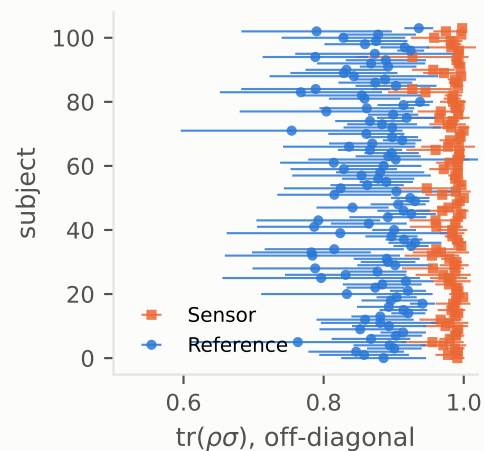
a

Sensor frame
cluster moves under A

A becomes a rotation U ;
unitary invariants are unchanged



Reference frame
spread, and invariant

b**c****d**

(a) EEG covariances cluster tightly about their mean, and the whole cluster moves under a nuisance congruence. Whitening by the mean spreads the states and reduces the nuisance to a rotation, which the quantum invariants do not see. (b) That invariance verified numerically. (c, d) The measured consequence on real EEG, $n = 104$ subjects.